

# Hamiltonian and time evolution in quantum system

*(Evolution in quantum systems)*

Classically, a system evolves because forces act upon it, positions and velocity changes under the effect of these forces. **But in quantum mechanics, what exactly evolves?**

A quantum state is a vector in Hilbert space. The fundamental object that describes a quantum system is the state vector  $|\psi(t)\rangle \in H$ . This vector contains all the information about the system.

So when we talk about “time evolution” in quantum mechanics, we mean:

**Time evolution = the state vector changes with time.**

However, normalization must be preserved all along system evolution, so the total probability remain equal to 1 at all times.

$$\langle\psi(t)|\psi(t)\rangle = 1 \quad \forall t$$

If the evolution takes an initial state  $|\psi(0)\rangle$  to a later state  $|\psi(t)\rangle$  then for any two states  $|\phi\rangle$  &  $|\psi\rangle$ :

$$\langle\phi(0)|\psi(0)\rangle = \langle\phi(t)|\psi(t)\rangle$$

Measurement probabilities depend on inner products, means that inner product is preserved.

The only transformations on a Hilbert space that preserve all inner products are unitary operators.

Thus, the most general form of time evolution is:

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle \quad , \quad U(t)^\dagger U(t) = I$$

*This is the most fundamental and general statement about time evolution in quantum mechanics.*

Physics does not allow the state to “jump” discontinuously when time changes by an infinitesimal amount. So we require:

$$\lim_{\Delta t \rightarrow 0} U(\Delta t) = I$$

*“Small changes in time produce small changes in the state.”*

Time evolution must satisfy:

$U(t_1)U(t_2) = U(t_1 + t_2)$  because evolving for  $t_1$  and then for  $t_2$  is the same as evolving for  $t_1 + t_2$ . This is called a **one-parameter unitary group**.

For finite-dimensional systems (which includes qubits), any continuous unitary one-parameter group has a Hermitian generator: “Any strongly continuous one-parameter unitary group can be written as:  $U(t) = e^{-iGt}$ ”

where  $G$  is a unique self-adjoint operator.

$$U^\dagger U = I$$

Differentiate this condition at  $t = 0$ :

$$G^\dagger = G$$

That shows:

Hermiticity is required by probability conservation

With other words this theorem tells us that **unitarity + continuity + group structure** force the time evolution operator to be an exponential of some **Hermitian** operator.

Let's call that operator  $\frac{H}{\hbar} \rightarrow \boxed{U(t) = e^{-iHt/\hbar}}$ .

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**The operator  $H$  as generator of time evolution:**

By definition:  $U(t) = e^{-iHt/\hbar}$

Differentiate at  $t=0$ :

$$\frac{d}{dt}U(t)|_{t=0} = \frac{-i}{\hbar}H$$

So:

$$H = i\hbar \frac{d}{dt}U(t)|_{t=0}$$

This shows:

**The Hamiltonian is the generator of infinitesimal time evolution.**

*The Hamiltonian is not introduced as “energy” here. It is first introduced as the generator required by unitarity and continuity.*

Deriving the Schrödinger equation

Start from:

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle$$

Differentiate with respect to time:

$$\frac{d}{dt} |\psi(t)\rangle = -\frac{i}{\hbar} H |\psi(t)\rangle$$

Multiply both sides by  $i\hbar$ :

$$\boxed{i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle}$$

This is the **time-dependent Schrödinger equation**.

So, we've established that time evolution in quantum mechanics is unitary:  
 $U(t) = e^{-iHt/\hbar}$ .

But why does the generator of time translations correspond to the physical quantity we call energy?

**For one:** Noether's theorem states:

*"Every continuous symmetry corresponds to a conserved quantity."*

**For two :** The laws of nature do not depend on when you perform an experiment; This is called time translation invariance, and this symmetry must be represented in quantum mechanics by a unitary operator.

For time translation symmetry, the conserved quantity is **energy**.

In quantum mechanics, this means:

- the generator of time translations must be the operator whose expectation value is conserved,
- that operator is what we call the Hamiltonian .

This is not a definition — it is a consequence of symmetry.

If  $|E_n\rangle$  is an eigenstate of ,

$$H |E_n\rangle = E_n |E_n\rangle$$

then its time evolution is:

$$|E_n(t)\rangle = e^{-iE_n t/\hbar} |E_n\rangle$$

Each energy eigenstate rotates in the complex plane with **angular frequency**:

$$\omega_n = \frac{E_n}{\hbar}$$

**Relation with relative phase:**

If the state is a superposition:

$$|\psi(0)\rangle = c_1 |E_1\rangle + c_2 |E_2\rangle$$

at time t:

$$|\psi(t)\rangle = c_1 e^{-iE_1 t/\hbar} |E_1\rangle + c_2 e^{-iE_2 t/\hbar} |E_2\rangle$$

the relative phase is:

$$\Delta\phi(t) = \frac{(E_2 - E_1)t}{\hbar}$$

This relative phase changes in time because the two energies rotate at different frequencies.